

A Contragredient View of Conformal Placement and Steinitz Balancing

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July 2026

ROUGH DRAFT — NOT FOR DISSEMINATION

Abstract

We give the validity of split conformal prediction an exact contragredient form: the placement acceptance map is equivariant for the permutation action, and the Reynolds identity $\langle P^\vee f, v \rangle = \langle f, Pv \rangle$ reads its orbit average either as moving the evaluation functional or as averaging the induced acceptance vector over placements. A natural primal discrepancy construction under the same action is Steinitz balancing: ordering a zero-sum population so that every prefix sum stays bounded. Conformal placements form such a population, and an explicit ordering keeps running coverage within $1/(2t)$ of the orbit level k/N , exact on the full orbit. On the function–measure pairing, the measure-side counterpart of calibration is balanced sampling and exact cubature. A transitive distributional symmetry, exchangeability in the standard setting, turns the deterministic orbit average into the coverage probability of the designated test placement.

1 Introduction

Split conformal prediction validates its coverage claim by a symmetry argument over sample slots. This note gives that argument an exact dual form and identifies the deterministic construction that lives beside it under the same group action. The identity behind conformal validity is a contragredient statement for the permutation action on slots (Section 3). A natural primal discrepancy construction under that action is *Steinitz balancing*: ordering a finite zero-sum population so that every prefix sum stays uniformly bounded (Steinitz, 1913). Conformal placements are themselves a scalar Steinitz population (Section 5), and admit a sharper balanced ordering than the general lemma provides. On the function–measure pairing, the measure-side counterpart of calibration is balanced sampling and exact cubature (Section 6). Section 7 records how the same two-ingredient structure appears in the dependent-data floor of Barber and Pananjady (2026), and Section 8 separates three rates that the shared algebra can blur. The group-symmetry view of predictive inference is itself established: Dobriban and Yu (2025) build predictive inference on general distributional group symmetries, of which exchangeability is the permutation case. What we believe is new here is narrower: the identification of the placement acceptance vector as a Steinitz population, and the balanced traversal it admits. Herding (Welling, 2009), whose $O(1/n)$ bookkeeping resembles the constructions here, can be viewed as a with-replacement relaxation of the finite balanced-permutation problem (Harvey and Samadi, 2014); it is confined to Remark 2 and a kernel question in Section 6.

2 Conformal calibration and placement

Fix $\alpha \in (0, 1)$, write $p = 1 - \alpha$, let $S_1, \dots, S_n$ be calibration scores and S_{n+1} a test score, almost surely distinct, and let $k = \lceil p(n+1) \rceil \leq n$. Write $N = n+1$. Split conformal declares coverage when $S_{n+1} \leq q_n$, with $q_n = S_{(k)}$ the k -th smallest calibration score.

Lemma 1 (Steering identity). *With $q_n = S_{(k)}$, $h_q(s) = \mathbf{1}\{s \leq q\}$, and the scores distinct,*

$$\sum_{i=1}^n (h_{q_n}(S_i) - p) = k - pn \in [p, 1+p).$$

Proof. Exactly k calibration scores satisfy $S_i \leq S_{(k)}$; and $p(n+1) \leq k < p(n+1) + 1$. $\square$

The material of this section is standard (Papadopoulos et al., 2002; Vovk et al., 2005; Lei et al., 2018).

Proposition 1 (Counting identity). *Let $s_1, \dots, s_N$ be distinct reals and, for $i \in \{1, \dots, N\}$, call placement i covered if s_i is at most the k -th smallest of $\{s_j\}_{j \neq i}$, writing $\chi_i(s) = \mathbf{1}\{s_i \leq Q_k(s_{-i})\}$ for the indicator. Then $\chi_i(s) = 1$ iff the rank of s_i among all N values is at most k , and consequently $\sum_{i=1}^N \chi_i(s) = k$.*

Proof. Placement i is covered iff at most $k-1$ of the other values are smaller than s_i , iff its overall rank is at most k ; ranks are a permutation of $\{1, \dots, N\}$. $\square$

Corollary 1 (Marginal validity). *If $(S_1, \dots, S_N)$ is exchangeable with almost surely distinct components, then $\mathbb{P}\{S_N \leq q_n\} = k/N \geq 1 - \alpha$.*

Proof. The coverage event is the event that placement N is covered. By exchangeability every placement indicator has the same expectation; average Proposition 1. $\square$

The control-theoretic reading of the calibration step is established in the online literature: the level update of adaptive conformal inference (Gibbs and Candès, 2021) is integral feedback on realized miscoverage, and Angelopoulos et al. (2023) name the identification. There the controlled quantity is a time average along a sequence; Lemma 1 is the batch counterpart, a single dead-beat correction.

3 The contragredient identity

Let $V = \mathbb{R}^N$, with $\mathfrak{S}_N$ acting by permuting slots, $(\rho(\pi)v)_j = v_{\pi^{-1}(j)}$, and let $\rho^\vee(\pi)f = f \circ \rho(\pi^{-1})$ be the contragredient action on the dual space V^* (for the dual representation and the averaging operator see, e.g., Serre, 1977). The coordinate functionals e_i^* and the prefix-sum functionals $L_t = \sum_{j \leq t} e_j^*$ are elements of V^* , and one has, exactly,

$$L_t(\rho(\pi)v) = (\rho^\vee(\pi^{-1})L_t)(v) : \tag{1}$$

permute the points holding the prefix functional fixed, or hold the points fixed and permute the functional. Averaging over the group, let $P = \frac{1}{N!} \sum_{\pi} \rho(\pi)$ be the Reynolds projector and $P^\vee = \frac{1}{N!} \sum_{\pi} \rho^\vee(\pi)$ its dual, so that

$$\langle P^\vee f, v \rangle = \langle f, Pv \rangle \quad \text{for all } f \in V^*, v \in V. \tag{2}$$

For the permutation representation, $Pv = \bar{v} \mathbf{1}$ and $P^\vee e_N^* = \frac{1}{N} \mathbf{1}^*$.

The object the projector acts on is not the score vector but the acceptance vector. Write $\mathbb{R}_{\neq}^N$ for the score vectors with distinct entries and $\chi : \mathbb{R}_{\neq}^N \rightarrow \{0, 1\}^N \subset V$ for the placement-acceptance map of Proposition 1. Since $\chi_i(s)$ depends only on the rank of s_i among the entries of s , the map is equivariant:

$$\chi(\rho(\pi)s) = \rho(\pi)\chi(s) \quad \text{for all } \pi \in \mathfrak{S}_N, s \in \mathbb{R}_{\neq}^N. \quad (3)$$

The distinction matters because χ is nonlinear: $Ps = \bar{s} \mathbf{1}$ destroys the rank information, and $\chi(Ps) \neq P\chi(s)$ in general. What is averaged along the orbit is the acceptance vector the scores induce, not the scores themselves.

Theorem 1 (Orbit-contragredient form of conformal validity). *For every $s \in \mathbb{R}_{\neq}^N$,*

$$\frac{1}{N!} \sum_{\pi \in \mathfrak{S}_N} \chi_N(\rho(\pi)s) = \langle e_N^*, P\chi(s) \rangle = \langle P^\vee e_N^*, \chi(s) \rangle = \frac{1}{N} \sum_{i=1}^N \chi_i(s) = \frac{k}{N}.$$

Consequently, whenever the law of S is invariant under a subgroup of $\mathfrak{S}_N$ acting transitively on the slots, in particular under exchangeability, the designated test placement has coverage probability exactly k/N .

Proof. The first equality is (3) and linearity of e_N^* ; the second is (2); the third is $P^\vee e_N^* = \frac{1}{N} \mathbf{1}^*$; the fourth is Proposition 1. Under a transitive invariance the indicators $\chi_i(S)$ share one expectation, which the orbit average then pins to k/N . $\square$

Define also the centered placement residual vector $r(s) = (\chi_i(s) - k/N)_{i \leq N}$; Proposition 1 says $\langle \mathbf{1}^*, r(s) \rangle = 0$, i.e.

$$\langle P^\vee e_N^*, r(s) \rangle = 0 \quad \text{for every distinct } s. \quad (4)$$

The identity is exact and requires no analogy; it is the defining identity of the dual representation, applied to the acceptance vector. It does not remove the distributional assumption: Theorem 1 is a statement about the whole orbit, and a transitive symmetry of the law, exchangeability being the standard case, is what makes the designated placement inherit the orbit average. The systematic development of predictive inference from distributional group symmetries is due to Dobriban and Yu (2025).

4 Steinitz balancing

A natural primal discrepancy construction under the same $\mathfrak{S}_N$ action is the ordering of a finite zero-sum population. Let E be a d -dimensional normed vector space; finite dimensionality is essential, as no constant independent of N is available in general otherwise. If $z_1, \dots, z_N \in E$ satisfy $\sum_i z_i = 0$ and $\|z_i\| \leq 1$, the Steinitz lemma (Steinitz, 1913) supplies a permutation π with every prefix bounded, $\|\sum_{j \leq t} z_{\pi(j)}\| \leq C_E$ for all t , with C_E depending on the space and norm but not on N (a bound linear in d holds for arbitrary norms; Grinberg and Sevast'yanov, 1980); hence running averages within C_E/t of zero. This is the bounded-ledger $O(1/t)$ phenomenon, obtained by an element of the same permutation group that Section 3 acts with.

Two precisions. First, the vector-valued setting scalarizes cleanly: on $W = E^N$ define, for each $\lambda \in E^*$, the functional $L_{t,\lambda}(z) = \lambda(\sum_{j \leq t} z_j) \in W^*$; then (1) holds verbatim for $L_{t,\lambda}$, and $\|\sum_{j \leq t} z_{\pi(j)}\| = \sup_{\|\lambda\|_* \leq 1} L_{t,\lambda}(\rho(\pi^{-1})z)$ recovers the norm bound. Second, Steinitz balancing is not itself the contragredient of conformal averaging. Every permutation, balanced or not, satisfies

the transport identity (1); the Reynolds average gives the exact orbit identity of Theorem 1; what Steinitz contributes is the choice of one π , data-dependent, that makes the whole nested family of prefix functionals small at once.

For the relation to herding, see Remark 2.

5 Balanced conformal placements

Conformal placements are themselves a scalar Steinitz population: by Proposition 1 the residuals $r_i(s) = \chi_i(s) - k/N$ satisfy $\sum_i r_i(s) = 0$ and $|r_i(s)| \leq 1$. Because they take only two values, the ordering can be made sharper than the general lemma provides. The balancing step is a statement about binary vectors, so we state it that way; the conformal content is that Proposition 1 produces such a vector.

Lemma 2 (Balanced ordering of a binary population). *Let $b \in \{0, 1\}^N$ with $\sum_i b_i = k$. There is a permutation $\pi \in \mathfrak{S}_N$ such that, simultaneously for every $t = 1, \dots, N$, $|\sum_{j \leq t} b_{\pi(j)} - tk/N| \leq \frac{1}{2}$.*

Proof. Define the balanced binary word $w_t = \lfloor tk/N + \frac{1}{2} \rfloor - \lfloor (t-1)k/N + \frac{1}{2} \rfloor$. Since $0 \leq k/N \leq 1$, each $w_t \in \{0, 1\}$, and $\sum_{j \leq t} w_j = \lfloor tk/N + \frac{1}{2} \rfloor$, so $|\sum_{j \leq t} w_j - tk/N| \leq \frac{1}{2}$ for every t , with exactly k ones among $w_1, \dots, w_N$. Send the indices with $b_i = 1$ to the positions with $w_t = 1$ and the rest to the rest. $\square$

Corollary 2 (Balanced placement ordering). *For every distinct score vector s there is a permutation $\pi \in \mathfrak{S}_N$ such that, simultaneously for every $t = 1, \dots, N$,*

$$\left| \frac{1}{t} \sum_{j=1}^t \chi_{\pi(j)}(s) - \frac{k}{N} \right| \leq \frac{1}{2t},$$

with equality to zero at $t = N$, and hence $|\frac{1}{t} \sum_{j \leq t} \chi_{\pi(j)}(s) - p| < \frac{1}{2t} + \frac{1}{N}$ against the nominal level p .

Proof. Apply Lemma 2 to $b = \chi(s)$, using Proposition 1; the second bound adds $|k/N - p| < 1/N$. $\square$

The lemma is one-dimensional balanced rounding; prefix bounds of this type are classical, going back to the chairman assignment problem (Tijdeman, 1980), and no specifically conformal structure enters once the count $\sum_i \chi_i(s) = k$ is known. The contribution is the identification: the conformal placement vector is an object to which balanced-word machinery applies, and the traversal it yields tracks the orbit level k/N , not the nominal p , exactly at the full orbit and within $1/(2t)$ before it. In the notation of Section 3 the two readings of the traversal are literal: $\langle L_t, \rho(\pi^{-1})r(s) \rangle = \langle \rho^\vee(\pi)L_t, r(s) \rangle$ — move the residual vector through the permutation holding the prefix test fixed, or hold the residuals fixed and move the nested family of prefix functionals.

The ordering is post hoc: constructing it requires knowing which placements are covered, hence the full score vector or its ranks. It is a deterministic traversal for orbit computations and repeated-placement audits, not an online algorithm, and it does not replace exchangeability in one-shot conformal prediction, where a single placement is designated as the test; only the distributional symmetry makes that placement uniform on the orbit.

6 Balanced sampling: the measure-side counterpart

If the pairing of interest is $\langle h, \nu \rangle = \int h d\nu$ rather than the permutation representation, calibration has a measure-side counterpart obtained by reversing the roles of the two arguments. Calibration

holds the empirical score measure ν_n fixed and chooses a test: find a threshold h_q with $\langle h_q, \nu_n \rangle = k/n$. The role-reversed problem holds tests $h_1, \dots, h_d$ and targets $m_1, \dots, m_d$ fixed and chooses a measure: find ν with $\langle h_j, \nu \rangle = m_j$. (This is an inverse moment problem under the pairing, not the adjoint of an operator: the adjoint of the moment map $T(\nu) = (\langle h_j, \nu \rangle)_{j \leq d}$ is $T^*a = \sum_j a_j h_j$, which lives on the function side.) Over a finite candidate population with $A_{ji} = h_j(x_i)$ the measure-side problem is the selection problem $Az = b$, $z \in \{0, 1\}^N$, which is precisely the balanced-sampling problem solved (exactly when geometry and integrality permit, approximately otherwise) by the cube method (Deville and Tillé, 2004). When arbitrary positive weights are allowed the problem changes and convex geometry answers it exactly: $\mu = \int \phi dP$ lies in the convex hull of $\phi(\mathcal{X})$, so Carathéodory supplies at most $d+1$ weighted points matching all d moments, the Tchakaloff side of exact cubature (Tchakaloff, 1957). In this formulation, balanced sampling and exact cubature are the measure-side counterpart of threshold calibration.

A designed calibration set sits inside this frame. If designed scores $S_1, \dots, S_n$ have empirical distribution function within Kolmogorov distance D of a continuous reference law $\bar{F}$, and the test score is drawn from $\bar{F}$ independently of the design, then $|\bar{F}(S_{(k)}) - k/(n+1)| \leq D + 1/(n+1)$ and $|\bar{F}(S_{(k)}) - p| < D + (1+p)/n$, deterministically in the design (the proof is $\hat{F}_n(S_{(k)}) = k/n$ plus Lemma 1). A quantile grid achieves $D \leq 1/(2n)$. This is design-based rather than distribution-free: it suppresses the $n^{-1/2}$ fluctuation of conditional coverage at the price of assuming the target law, and whether reproducing-kernel guarantees of kernel herding or kernel thinning (Chen et al., 2010; Dwivedi and Mackey, 2021) control the Kolmogorov discrepancy required here is left open.

7 The Barber–Pananjady floor

The two-ingredient structure of the preceding sections reappears in the dependent-data coverage floor of Barber and Pananjady (2026). For $S = (S_1, \dots, S_N)$ they define subvectors $\Delta_{k,\tau}^0(S)$ and $\Delta_{k,\tau}^1(S)$ of length $N - \tau$, deleting a block of τ entries while keeping S_N (respectively rotating S_k) in the final slot, and set

$$\Psi_{k,\tau}(S) = d_{\text{TV}}(\mathcal{L}(\Delta_{k,\tau}^0(S)), \mathcal{L}(\Delta_{k,\tau}^1(S))), \quad \bar{\Psi}_\tau = \frac{1}{N} \sum_{k=1}^N \Psi_{k,\tau}(S),$$

with $d_{\text{TV}}(P, Q) = \sup_A |P(A) - Q(A)|$. Their Theorem 5 gives, for a pretrained memoryless score,

$$\mathbb{P}\{S_N \leq q_n\} \geq 1 - \alpha - \min_{0 \leq \tau \leq n} \left\{ \frac{\tau}{N} + \bar{\Psi}_\tau \right\},$$

their Proposition 3, carried to memoryless scores by their Proposition 4, bounds $\bar{\Psi}_\tau \leq 2\beta(\tau)$ under stationary β -mixing, their Theorem 8 supplies a matching ceiling, and their Theorem 7 shows tightness up to the factor $(1 - \alpha)/4$.¹

The argument separates into an exact insertion–deletion stability statement for empirical quantile *levels* (deleting τ entries moves the rank of a fixed threshold by at most τ , so the level shifts by at most τ/N : order-statistic geometry, controlling levels rather than values, since empirical quantile values are not stable under small total-variation perturbations) and a total-variation comparison between switched laws, which is the definition of the switch coefficient. Both ingredients are linear in their perturbation sizes and no concentration step appears; a blocking argument (Oliveira et al., 2024) bounds the same quantity through a maximal inequality and pays $\sqrt{\tau/n}$.

¹Result numbering follows the ALT 2026 (PMLR volume 313) version of Barber and Pananjady (2026); the arXiv versions number the same results differently.

8 Three rates

The shared algebra can blur three quantities. The marginal coverage k/N of Corollary 1 is an exact symmetry identity; its $O(1/n)$ deviation from p is rank-grid rounding, not estimation error. The calibration-sample-conditional coverage $\mathbb{P}\{S_{n+1} \leq q_n \mid S_1, \dots, S_n\} = \bar{F}(q_n)$, defined when the test score is drawn from a continuous $\bar{F}$ independently of the calibration scores, fluctuates around nominal at the sampling scale: for $S_1, \dots, S_n$ i.i.d. from $\bar{F}$ one has exactly $\bar{F}(S_{(k)}) \sim \text{Beta}(k, n+1-k)$, with standard deviation $O(n^{-1/2})$ (Vovk et al., 2005). The design of Section 6 suppresses this at the price of assuming $\bar{F}$. Steinitz discrepancies are genuine approximation errors, $O(1/t)$ with constants from the Steinitz lemma. The ledgers match; the objects differ.

9 Remarks

Remark 1 (The pinball dual). Threshold calibration also has an exact convex dual. The conformal threshold $q_n = S_{(k)}$ is the empirical quantile at level $\gamma = k/n$, not generally at p , and it minimises the pinball loss (Koenker and Bassett, 1978) at that level, $\sum_i \rho_\gamma(S_i - q)$ with $\rho_\gamma(r) = \sup_{u \in [\gamma-1, \gamma]} ur$. Fenchel duality gives

$$\max \left\{ \sum_i u_i S_i : u_i \in [\gamma-1, \gamma], \sum_i u_i = 0 \right\},$$

and with distinct scores $u_i = \gamma - h_{q_n}(S_i)$ is feasible for every i , including at the threshold, since $\sum_i u_i = n\gamma - k = 0$. The dual optimum is a balanced signed measure on the scores: calibration’s exact optimization dual, though not itself a point-generation algorithm.

Remark 2 (Herding). Herding (Welling, 2009) generates points by the recursion $w_t = w_{t-1} + \mu - \phi(x_t)$, $x_t = \arg \max_x \langle w_{t-1}, \phi(x) \rangle$, and attains $O(1/n)$ feature-mean discrepancy conditional on bounded weights; the identification with Frank–Wolfe (Bach et al., 2012) shows the fast rate needs interiority-type conditions, with Chen et al. (2010) treating the kernel case. Harvey and Samadi (2014) introduce a Permutation Problem, each point used once, that is exactly the balanced-permutation problem of Section 4, and show that solving it solves herding over the finite population; generic herding runs with replacement, typically on nonfinite domains (Welling, 2009; Chen et al., 2010), and on the pairing of Section 6 it is one iterative approximate solver for balanced sampling. The quadrature triangle is older on that side: optimally-weighted herding is Bayesian quadrature (Huszár and Duvenaud, 2012), and Snell and Griffiths (2025) recently recast the conformal computation as Bayesian quadrature over the calibration scores.

Remark 3 (Limits). The passage from orbit identity to one-shot coverage is exchangeability’s, or its mixing surrogate’s in Section 7; no bookkeeping supplies it. Nor does control of thresholds and labels create information: it re-levels coverage while conditional sharpness remains a property of the scores (Cotton). Whether placement balancing can be aimed conditionally, selecting or reweighting calibration slots to balance the rank count given the input, is open; localized and weighted conformal methods are the nearest existing steps.

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